

Lung ROI Segmentation — SwinUNETR (2D) — sliding-window (256^2), native resolution

1. Task

Per-slice binary lung segmentation of axial CT slices at the native 1 mm grid. Same task as the earlier resize-recipe ROI models — the difference is the recipe: **256×256 patch training + 2D sliding-window inference at native resolution** instead of resizing each slice to 256×256 (which distorts the lung outline on non-square slices).

- **Input** (1, H, W) axial slice, intensity-normalised to [0, 1], native voxel grid (270–500 px per side in this corpus).
- **Training input** 256×256 patches via `RandCropByPosNegLabeld` (pos:neg = 1:1, 4 patches/slice); slices smaller than the patch are zero-padded to 256×256.
- **Inference** MONAI `sliding_window_inference`, ROI (256, 256), overlap = 0.5, Gaussian blending, sigmoid > 0.5.
- **Output** (1, H, W) binary lung mask.

2. Model

SwinUNETR (Swin transformer encoder + U-Net decoder), `feature_size`: 24 (small variant), 1-channel sigmoid head. Trainable parameters: **6,302,203**.

3. Data

`unified` split, slices with lung annotation (NSCLC-Radiomics + NLST): **365,014 train / 64,097 val / 45,751 test** slices. Patient-grouped, dataset-stratified split, seed 42.

4. Training

Setting	Value
Loss	Dice (sigmoid, squared-pred)
Optimizer	Adam, LR 1×10^{-4} , weight decay 1×10^{-5}
LR schedule	Cosine annealing, $T_{\text{max}} = 100$, $\eta_{\text{min}} = 1 \times 10^{-6}$
Batch size	4 slices × 4 patches = 16 patches / step
Samples per epoch	20 000 slices (random subsample)
Epochs	100
Mixed precision	bf16

Setting	Value
Best val Dice	0.9702

5. Results — held-out test split (45,751 slices)

Metric	Value
Dice / F1 (micro)	0.9834
Precision	0.9836
Recall	0.9832
mIoU	0.9826
IoU (lung)	0.9674

Per-slice Dice distribution

Statistic	Value
Mean	0.9612
Std	0.1258
Median	0.9898
p05	0.8692
p95	1.0000

Unlike the resize-recipe SegResNet baseline (per-slice p05 = 0.0), both SW-trained ROI models keep a usable Dice on the near-empty apex/base slices — the native-resolution patch recipe removes that failure mode.

6. Reproducibility

- Config `configs/roi_swin_sw.yaml`
- Checkpoint `checkpoints/roi_swin_sw/best_model.pth`
- Metrics JSON `results_eval/eval_roi_swin_sw_test.json`
- Command `python eval_roi.py --config configs/roi_swin_sw.yaml --ckpt checkpoints/roi_swin_sw/best_model.pth --split test`